

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

JULY 2023

Course Code: CIV102

Course: ENGINEERING MECHANICS

QP No. :UF020-2

Duration: 3 hours

Max. Marks:100

PART-A

Answer all the questions

10 x 2 = 20 Marks

1. The resultant of two concurrent forces is 1500 N and the angle between the forces is 90° . The resultant makes an angle of 36° with one of the forces. Find the magnitude of each force.
2. A force of magnitude 800 N makes 20° with the y-axis and its projection on the x-z plane makes 30° with the x-axis. Express the force in the vector form.
3. Four parallel forces of magnitudes 100 N, 150 N, 25 N and 200 N are shown in Fig.1. Determine a single force and couple at A.

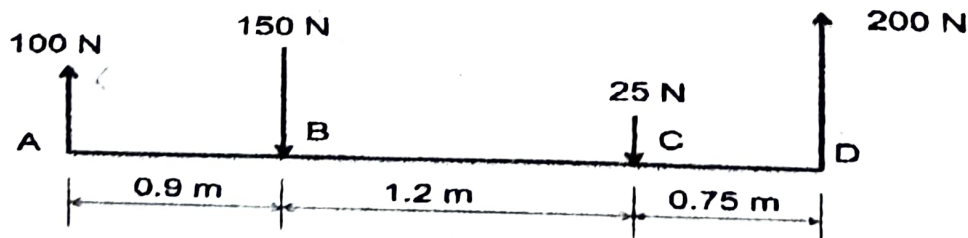


Fig.1

4. Two composite solid objects are made up of the same shape. In which one of the solids is made with two distinct dense materials

and another with homogeneous material. Will these two solid objects have a common centre of gravity? Justify your answers.

5. For cement manufacturing plant comprises the hopper bin section as shown in Fig.2. Calculate the inner surface area of the hopper bin.

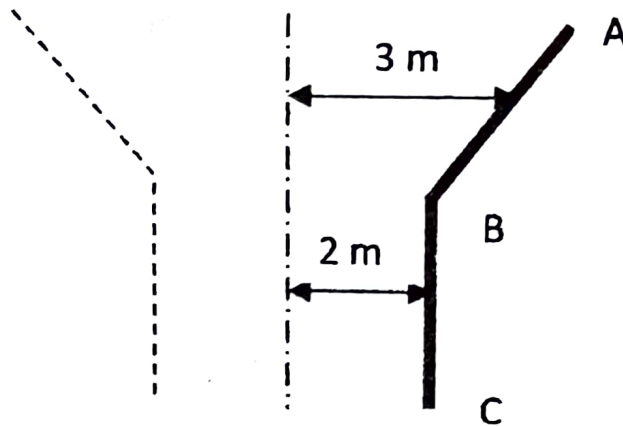


Fig.2

Length of section line segments: $AB = 10 \text{ m}$ and $BC = 8 \text{ m}$

6. For the solid shaft of a turbine generating power has the circumference of the shaft = 942.477 mm. Determine the moment of inertia of the shaft about its axis of rotation.
7. Outline the laws of dry friction and their limitations.
8. Illustrate with figures, the mechanism involved in the study of simple block friction subjected to externally applied force 'P' and friction force 'F'.
9. The velocity of a particle moving in a straight line is given by the expression: $v = t^3 - t^2 - 2t + 2$. The particle is found to be at a distance of 4 m from station A after 2 seconds. Determine acceleration after 4 seconds.
10. A glass marble, whose weight is 0.2 N, falls from a height of 10 m and rebounds to a height of 8 m. Find the impulse and the average force between the marble and the floor if the time during which they are in contact is $1/10^{\text{th}}$ of a second.

PART-B

Answer all the questions

4 x 15 = 60 Marks

11. A non-concurrent system of forces is acting on a plane ABCD as shown in Fig.3. Determine the resultant of the system and location of the same with reference to point A.

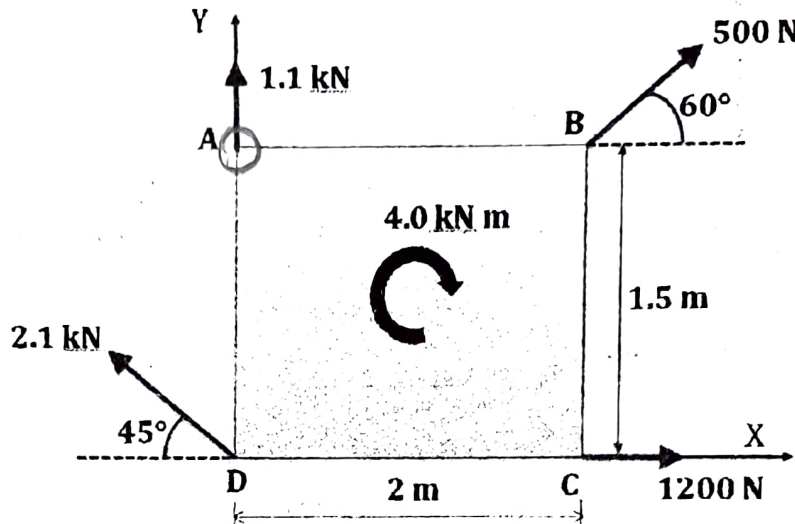


Fig.3

(OR)

12. The loads transferred by four columns to the concrete mat foundation are as shown in Fig.4. Determine the resultant and the point of application.

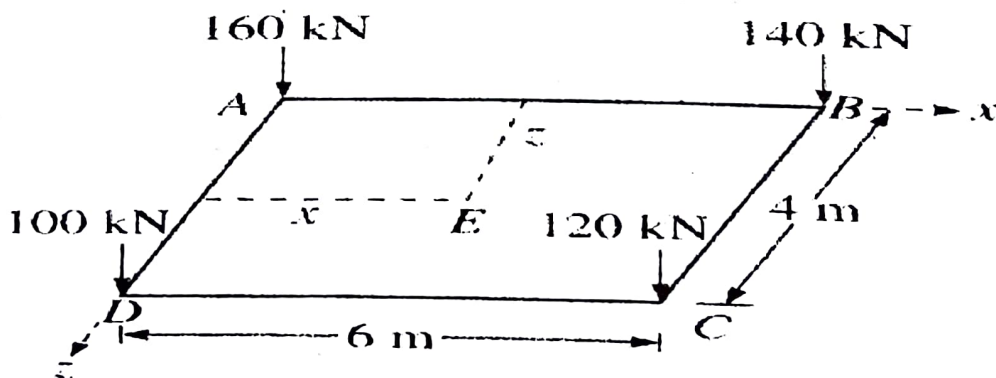


Fig.4

13. Locate the coordinates of the centroid of the plane area shown in Fig.5 with reference to the axes OX & OY shown. Take $a = 40$ mm.

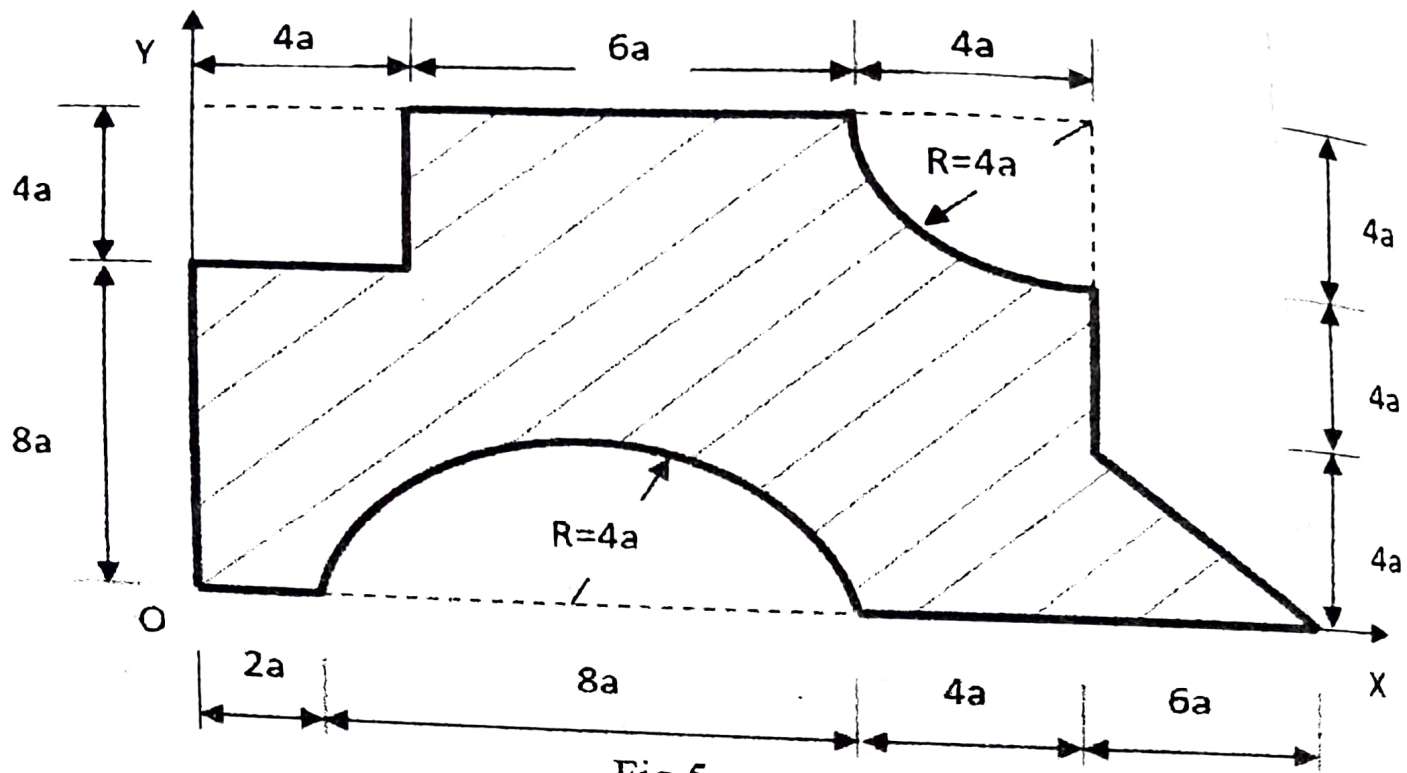


Fig.5

(OR)

14. Determine the moment of inertia of the section about the x - x and y - y axis passing through the centroid as shown in Fig. 6

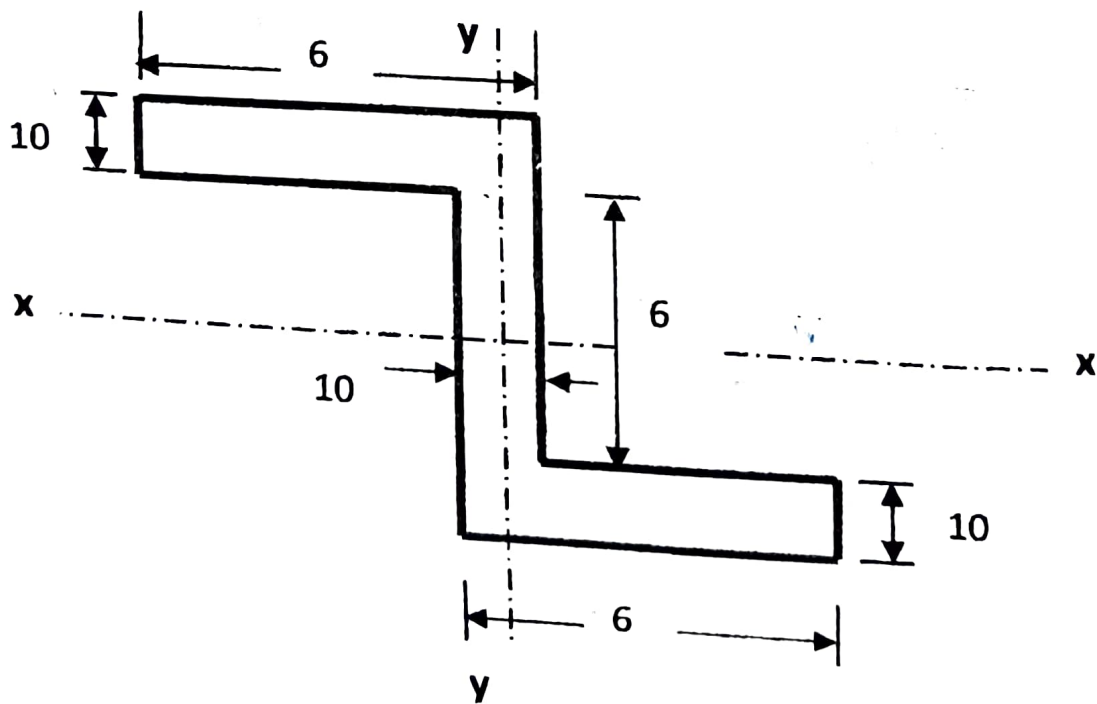


Fig.6

All Dimensions are in mm.

15. What is the value of P in the system shown in Fig. 7 to cause the motion to impend? Assume the pulley is smooth and the coefficient of friction between the other contact surfaces is 0.20.

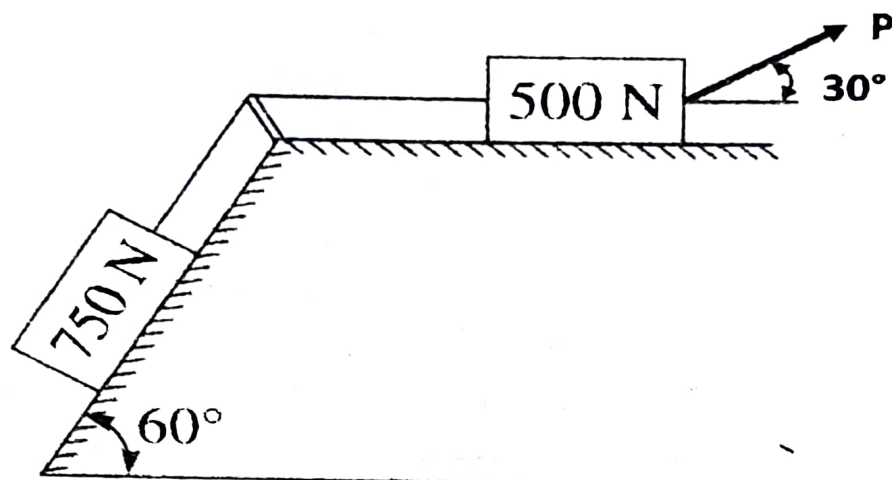


Fig.7

(OR)

16. In Fig.8 the coefficient of friction on all contact surfaces (including those between the pulley and the rope) is 0.30. Determine the minimum weight ' W ' to prevent the downward motion of the 1000 N body. Consider the angle of wrap = 180° .

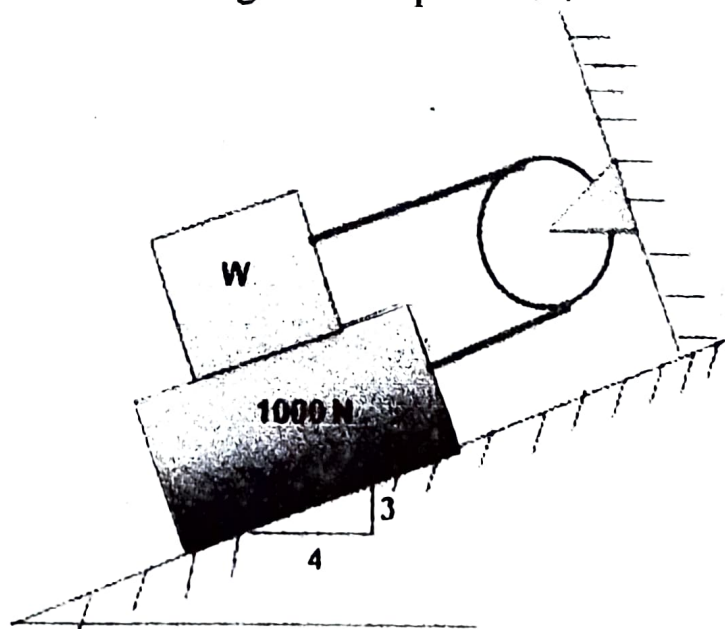


Fig.8

17. A soldier fires a bullet at an angle of 30° (upward from horizontal) from his position on a hill to strike a target which is 61 m lower than

the position of the soldier. The initial velocity of the bullet is 91.5 m/s. Calculate the following:

- The maximum height to which the bullet will rise above the horizontal.
- The actual velocity with which it will hit the target.
- The total time required for the flight of the bullet.

(OR)

18. Two blocks 40 kg and 24 kg are connected by inextensible strings and supported as shown in Fig.9. Determine the tension in the string and time taken by block 24 kg to attain a velocity of 2 m/sec from rest. Consider the pulley as frictionless and weightless. Take the coefficient of friction between block 40 kg and plane as 0.25.

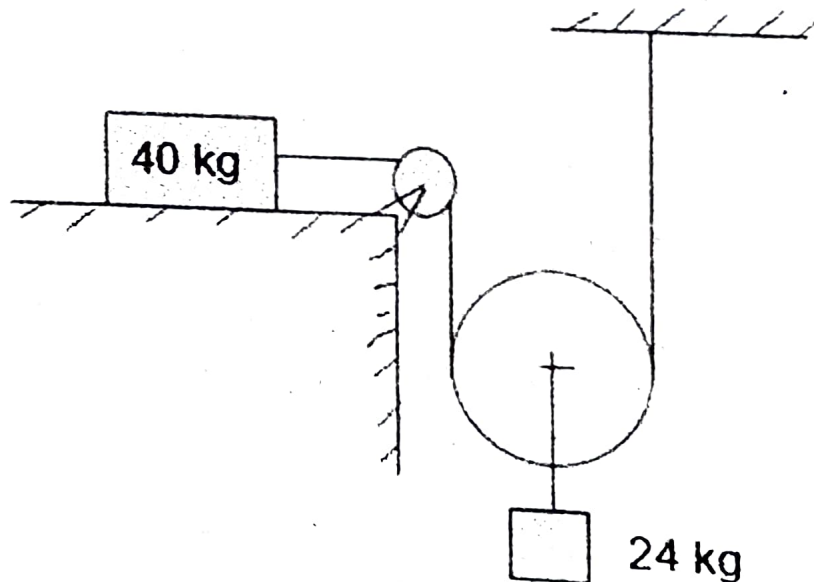


Fig.9

PART-C

Answer the following

1 x 20 = 20 Marks

- Forces 30 kN, 40 kN, 50 kN, and 60 kN are concurrent at $O(1,2,3)$ and are directed through $M(6,3,-2)$, $N(-4,-2,5)$, $P(-3,2,4)$, and $Q(4,-3,6)$ respectively. Determine the resultant of the system. (10)
- Determine the tension in the string and acceleration of blocks A & B weighing 1500 N & 500 N connected by an inextensible string as shown in Fig.10. Assume pulleys as frictionless &

weightless.

(10)

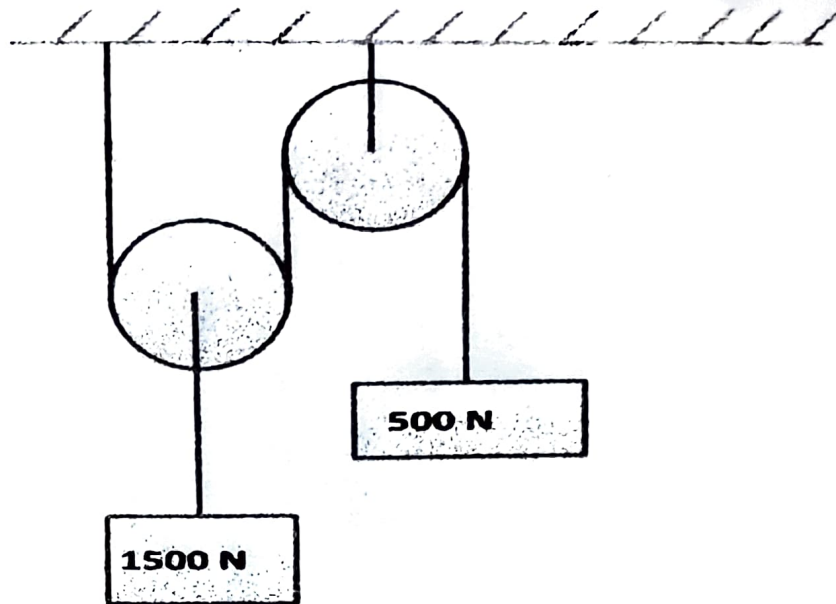


Fig.10
